Aryan Sood

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Education

Indian Institute of Technology Roorkee — **CGPA: 8.367/10 (3.567/4)***

Bachelor of Technology in Electrical Engineering — *Aug. 2024 – 2028*

* *US scale conversion calculated via Scholaro methodology.*

Experience

Auric AI Labs — **March 2026 – May 2026**

Computer Vision Intern

- Worked on generating 2D renders from 3D meshes of military objects using Blenderproc and further enhancing their quality using EO data.
- Created a Semantic Suggestion Pipeline via re-utilizing the methodology proposed in the paper Seeing The Unseen for EO data, and further implementing deterministic placement rules for military objects in satellite data.

Publications

More Than a Quick Glance: Overcoming the Greedy Bias in KV-Cache Compression — **ICLR 2026**

Preprint | **GitHub**

- *Authors:* **Aryan Sood**, Tanvi Sharma, Vansh Agrawal
- Developed LASER-KV, a KV-cache compression method achieving up to **10% higher accuracy at 128k context** over greedy pruning baselines, via block-wise accumulation and **Exact-LSH sampling**

Dropping the Anchor: Statistical Summarization for Distributed Systems via Pulsar Attention — **Under Review**

Preprint | **GitHub**

- *Authors:* **Aryan Sood**, Shantanu Acharya
- Collaborated with a researcher at NVIDIA, to further develop StarAttention (a framework for distributed systems). Pulsar Attention enables cross block communication using lightweight summaries which lead to a **4.97%** improvement at long context over the full attention baseline.

Don't Look Up (Every Token): Escaping Quadratic Complexity via Patterns and Algorithms — **ICLR 2026**

Blog

- *Authors:* **Aryan Sood**, Tanvi Sharma, Vansh Agrawal
- Surveyed sparsity-based efficiency methods for transformers and organized them into a unified taxonomy distinguishing **static structural patterns** from **dynamic, attention-driven** approaches

Projects

Sparse Attention Pre-Selection | GitHub — ***Sep. 2025***

- Designed a neural framework that reduces autoregressive inference overhead in LLMs by dynamically selecting relevant context tokens per query
- Incorporated a layer-wise sparse attention mask by adapting TidalDecode principles, substantially reducing per-step computational complexity

Adaptive Momentum-based PD Controller for Diffusion LLMs | GitHub — ***Dec. 2025***

- Built a training-free monotonic masking framework for Diffusion LLMs, applying momentum to project discrete token spaces into continuous domains
- Implemented a momentum heuristic that drives embeddings off-manifold, achieving earlier convergence on GSM8K and characterizing hallucination lock-in on MATH-500

Story Generation with Images | GitHub — ***Dec. 2024***

- Built a multimodal pipeline combining Stable Diffusion for image synthesis with a LoRA-finetuned OPT-1.3B model for coherent narrative generation
- Designed feedback loops with targeted penalties and negative prompting to enforce semantic alignment between generated images and text

Image Classification on an Adversarially Attacked Dataset | GitHub — ***Jan. 2025***

- Constructed a robust classification pipeline using an ensemble of CLIP models to handle adversarially perturbed inputs
- Improved robustness by incorporating diverse prompt templates and test-time augmentation (TTA) strategies

GNSS Anti-Spoofing Detection | GitHub — ***Mar. 2026***

- Built a Transformer encoder with 8-channel early fusion over 14 features per satellite, detecting GPS spoofing by learning cross-satellite signal inconsistencies; achieved **Val F1 of 0.9798** across 5-fold CV
- Engineered paper-inspired features (CNO spatial/temporal std, EML discriminator) capturing the physical signature of single-source spoofing; trained with AdamW, BCELoss, and per-fold early stopping

Achievements

- **All India Rank 1519, JEE Advanced 2024** | *Top exam for IIT admission, taken by 150k+ students nationally*
- **All India Rank 2912, JEE Main 2024** | *Qualifying exam for JEE Advanced, taken by 1M+ students nationally*
- **Top 20 of 1000+ Projects** | *Recognized for exceptional technical implementation among IIT Roorkee peers*
- **2nd Place, PixelPlay 2025** | *Week-long computer vision hackathon among top AI talent at IIT Roorkee*

Extracurricular

Vision and Language Group, IIT Roorkee — **Feb. 2025 – Present**

Core Member | **VLG Website** | **GitHub** — *IIT Roorkee*

- Active member of a deep learning research group focused on computer vision and language modeling, contributing to collaborative projects and technical discussions